TECHNISCHE UNIVERSITÄT DARMSTADT

PHYSICS-INSPIRED OPTIMIZATION OF THE **QUADRATIC ASSIGNMENT PROBLEM** ON GPU

Moritz Klein



Graph Matching for
Human Brain Connectomes

August 6th, 2026

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1 The Problem

QAP, QUBO Bottleneck

2 Algorithms

Overview, Deep-Dives, ASSA.jl

3 Results

QAPLIB, Speed, Brain Matching

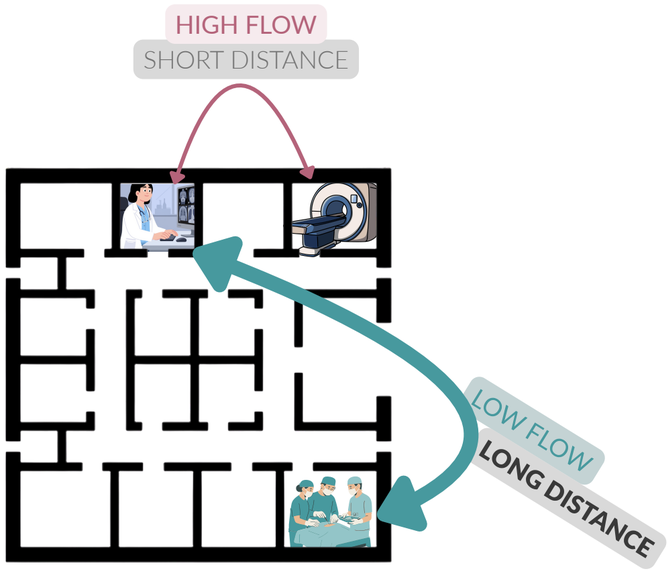
4 Conclusions

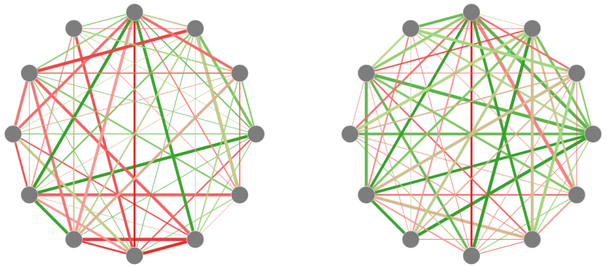
5 Future

Ideas, Teasers

THE PROBLEM

Hospital Design





QAP

$$\underset{\pi}{\operatorname{argmin}} \quad z(\pi) = \sum_{i}^{n}\sum_{j}^{n} A_{ij}B_{\pi_i\pi_j}$$
$$\simeq \operatorname{tr}(AXBX^T)$$

A = FLOW MATRIX

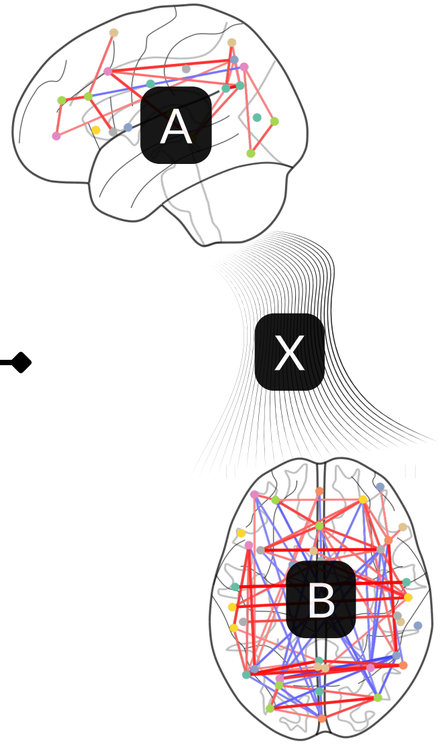
B = DISTANCE MATRIX

X = PERMUTATION
(One-hot)

NP-Hard

- Circuit Design
- Facility layout
- Keyboard layout
- Traveling Salesman Problem
- ...

Graph Matching



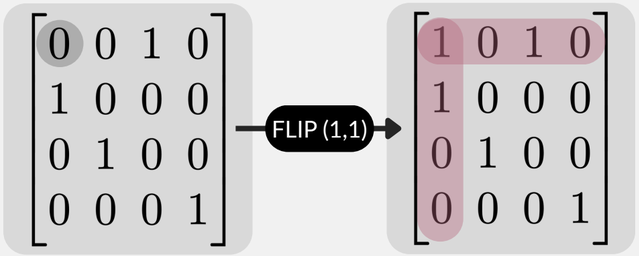
$$\underset{X \in P}{\operatorname{argmin}} \|A - XBX^T\|_F^2$$

0, 1

QUBO

$$Q = A \otimes B + \lambda P$$

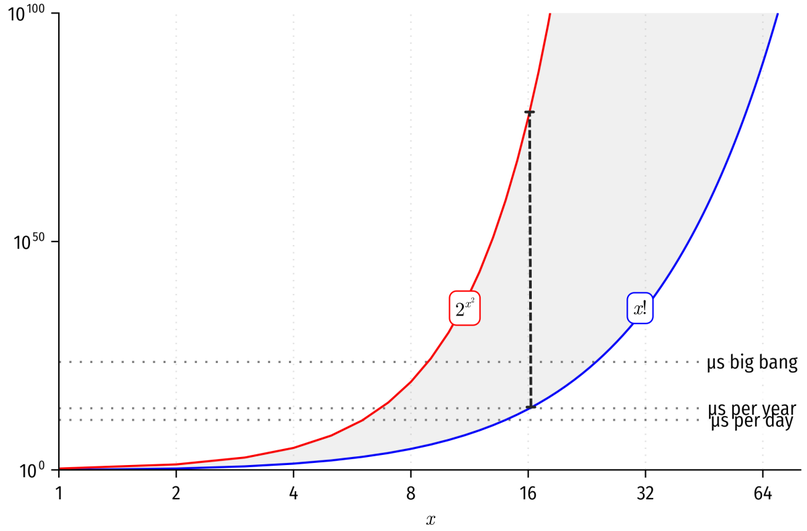
$$\underset{x \in P}{\operatorname{argmin}} \quad x^T Q x$$

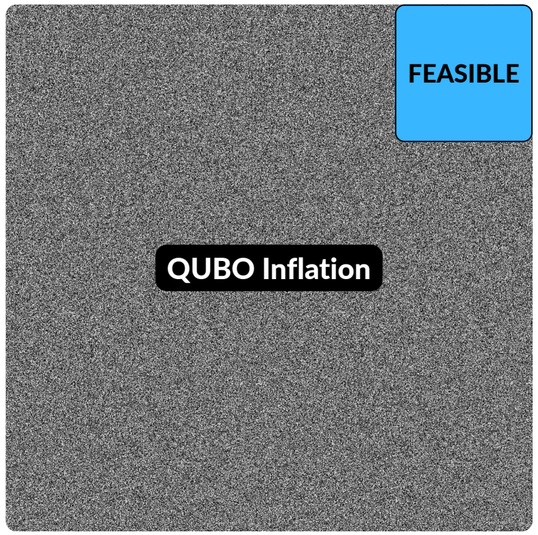


- Encode penalties to enforce feasibility
  - Big-M Problem
- Embedding
- Current hardware is limited to $n \approx 15$

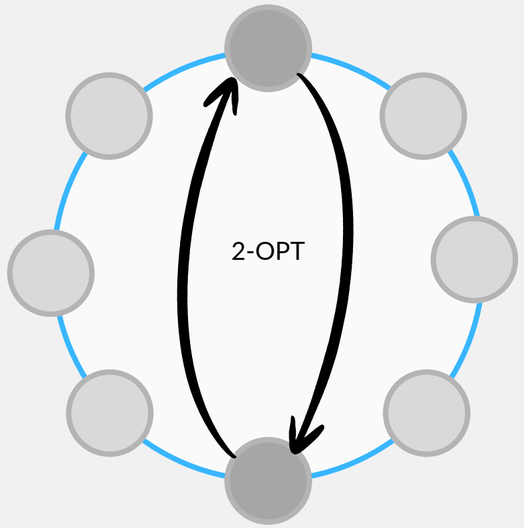
2, 3, 4, 5

$$2^{n^2} \gg n!$$





Native



$$[3, 1, 2, 4] \xrightarrow{\text{2OPT(1,3)}} [2, 1, 3, 4]$$

- Feasible by design
- ΔE: O(n) → with cache O(1) – *expensive to update*

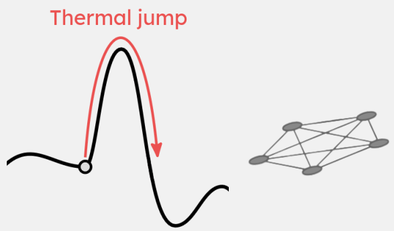
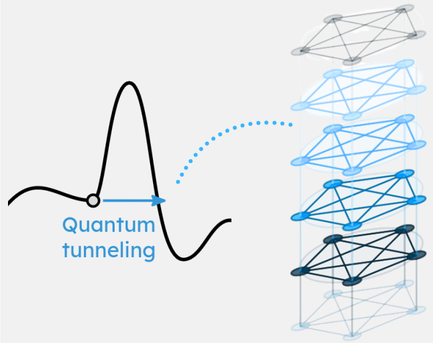
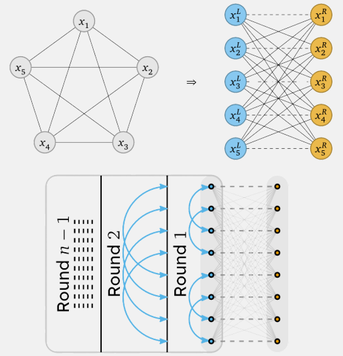
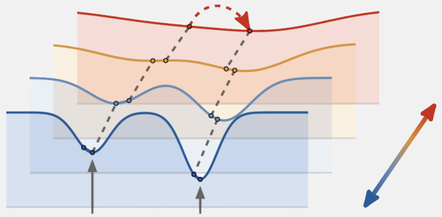
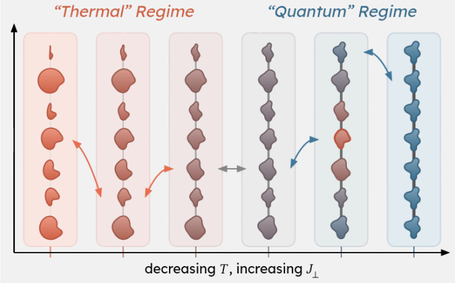
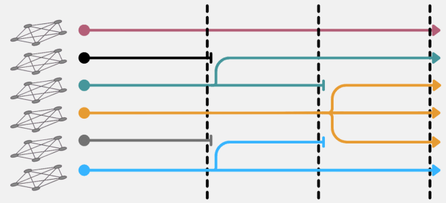
ALGORITHMS

AND **DEEP DIVES**

HEURISTICS OVERVIEW

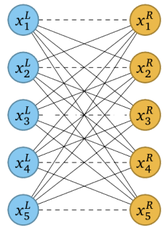
PRIMARY MECHANISM

ARCHITECTURE

	Thermal Fluctation	Quantum Tunneling	Parallelism & Survival of the fittest
Trajectory	Simulated Annealing **(SA)** 	Simulated Quantum Annealing **(SQA)** 	Momentum Annealing **(MA)** 
Population	Parallel Tempering **(PT)** 	**PT-SQA** 	Population Annealing **(PA)** 

unexplored for the QAP

4, 6, 7, 8, 9



Variables are duplicated into left and right copies that cross-interact.

Swap everything with everything

Use **Round-Robin tournaments** to generate all distinct matches for the 2-opt!



Side-Condition Cache Breakthrough

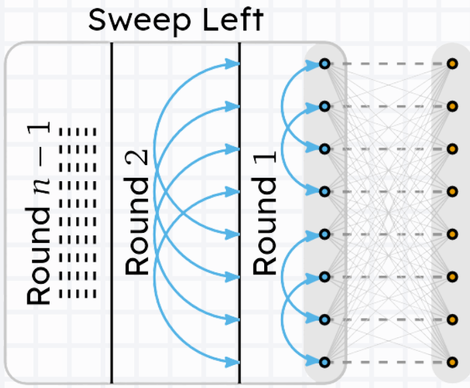
Same side: Fast $O(1)$ cache lookup

Switch side: Rebuild cache in $O(n^2)$

Key Insight: Because one side is frozen, the cache remains valid for an entire local sweep. The algorithm rebuilds the cache only on side-switch.

Round Robin Schedule

×	1	2	3	4	5	6	7	8	9	10	11	12	13
1	1	2	3	4	5	6	7	8	9	10	11	12	13
2	2	3	4	5	6	7	8	9	10	11	12	13	1
3	3	4	5	6	7	8	9	10	11	12	13	1	2
4	4	5	6	7	8	9	10	11	12	13	1	2	3
5	5	6	7	8	9	10	11	12	13	1	2	3	4
6	6	7	8	9	10	11	12	13	1	2	3	4	5
7	7	8	9	10	11	12	13	1	2	3	4	5	6
8	8	9	10	11	12	13	1	2	3	4	5	6	7
9	9	10	11	12	13	1	2	3	4	5	6	7	8
10	10	11	12	13	1	2	3	4	5	6	7	8	9
11	11	12	13	1	2	3	4	5	6	7	8	9	10
12	12	13	1	2	3	4	5	6	7	8	9	10	11
13	13	1	2	3	4	5	6	7	8	9	10	11	12



$$\boldsymbol{H}^L = \boldsymbol{A}\boldsymbol{X}_R\boldsymbol{B}^T = \boldsymbol{A}\boldsymbol{B}^T_{*,\pi_R}$$

$$\Delta z_{L(\pi^L,i,j)} = \boldsymbol{H}^L_{i,\pi^L_j} + \boldsymbol{H}^L_{j,\pi^L_i} - \boldsymbol{H}^L_{i,\pi^L_i} - \boldsymbol{H}^L_{j,\pi^L_j}$$

$$\boldsymbol{H}^R = \boldsymbol{A}^T\boldsymbol{X}_L\boldsymbol{B} = \boldsymbol{A}^T\boldsymbol{B}_{\pi_L,*}$$

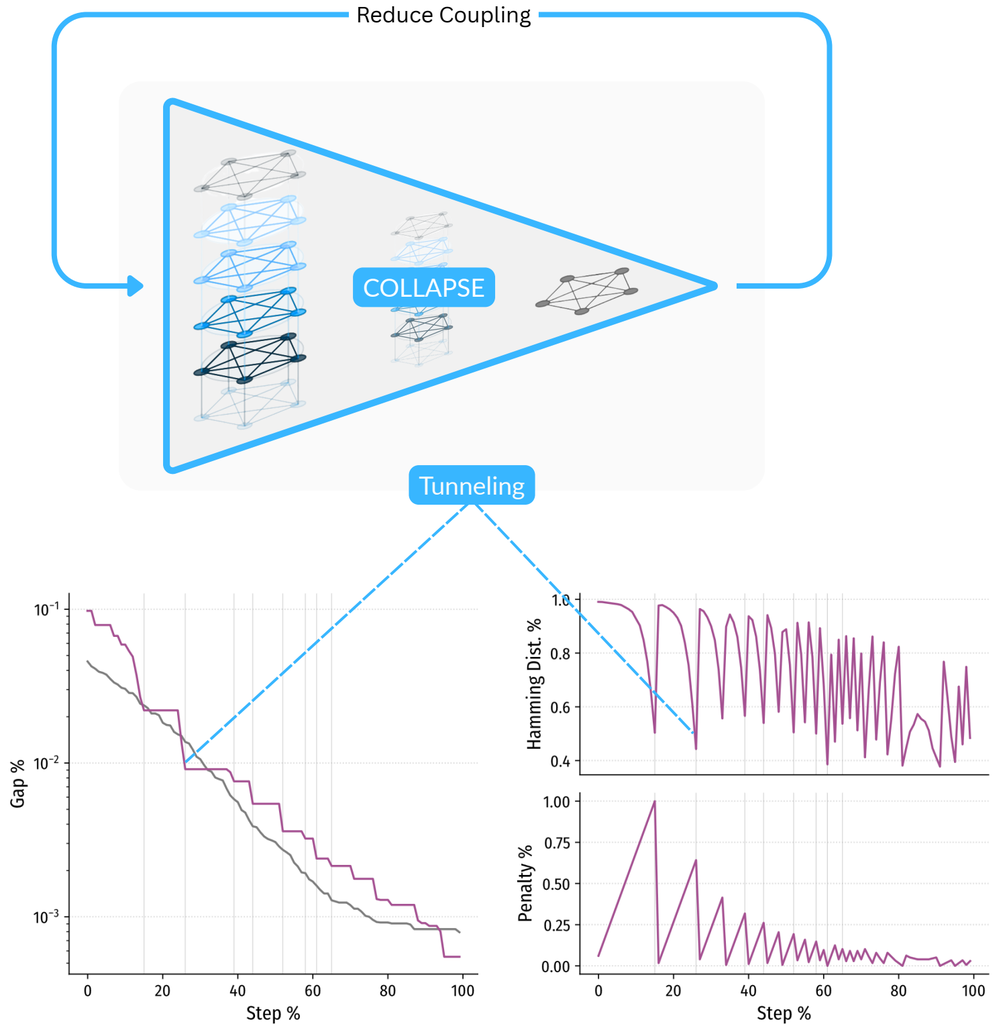
$$\Delta z_{R(\pi^R,i,j)} = \boldsymbol{H}^R_{i,\pi^R_j} + \boldsymbol{H}^R_{j,\pi^R_i} - \boldsymbol{H}^R_{i,\pi^R_i} - \boldsymbol{H}^R_{j,\pi^R_j}$$

Binary spin agreements are abandoned. Instead we use slice-wise **Hamming distances** between adjacent Trotter slices to dictate the **collapse**.

$$\mathcal{H}_{d+1} = \sum_{k}^{P} \left[\sum_{\{i,j\} \in \mathcal{V}}^{N} \boldsymbol{J}_{ij} \boldsymbol{s}_i^{[k]} \boldsymbol{s}_j^{[k]} + \sum_{i}^{N} \boldsymbol{h}_i \boldsymbol{s}_i^{[k]} + J_\perp \sum_{i}^{N} \boldsymbol{s}_i^{[k]} \boldsymbol{s}_i^{[k+1]} \right]$$

$$\overset{QAP}{\Longrightarrow} \sum_{k=1}^{P} \left[z\left(\pi^{[k]}\right) + J_\perp \sum_{i=1}^{N} 1 - \delta\left(\pi_i^{[k]}, \pi_i^{[k+1]}\right) \right]$$

Slice-wise Hamming distance

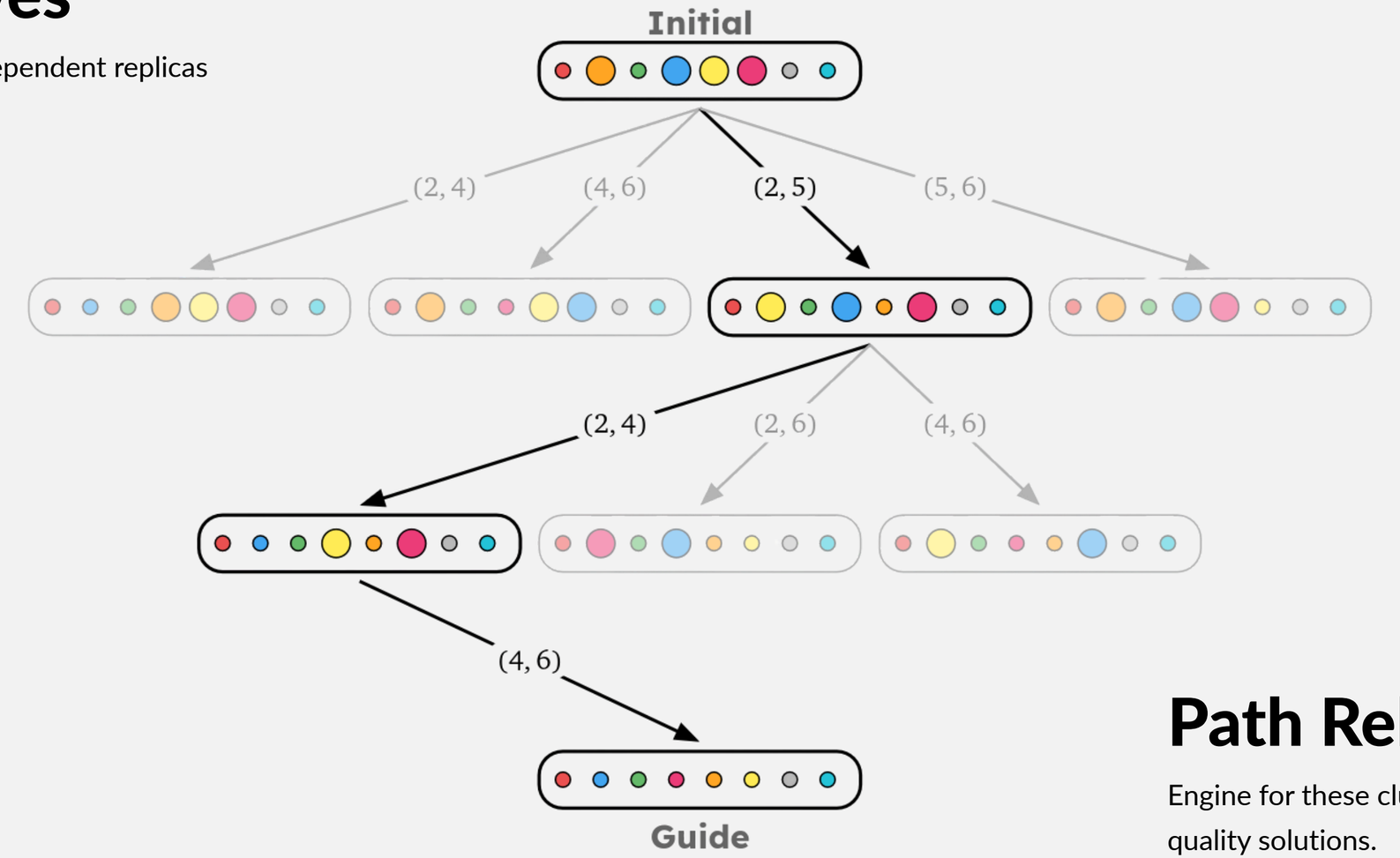


But still only one temperature direction → PT-SQA!

6, 15

Cluster Moves

Crossover between two independent replicas



Path Relinking

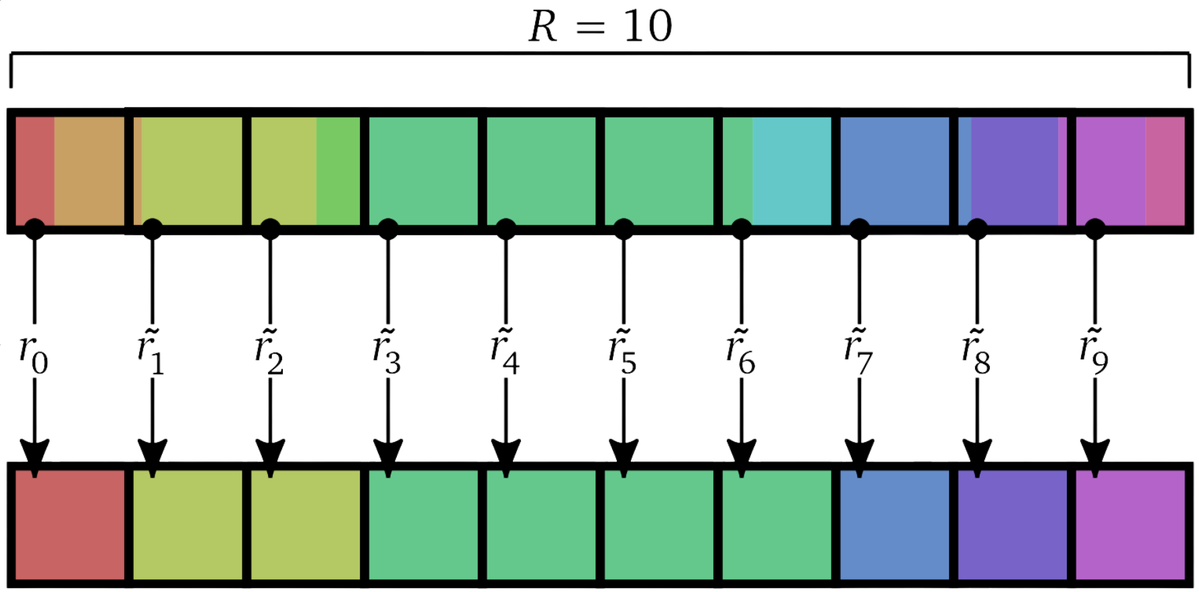
Engine for these cluster moves to combine two high quality solutions.

FITNESS

$$r_j = \frac{e^{-(\beta_i - \beta_{i-1})E_j}}{Q_i}$$

NORMALIZER

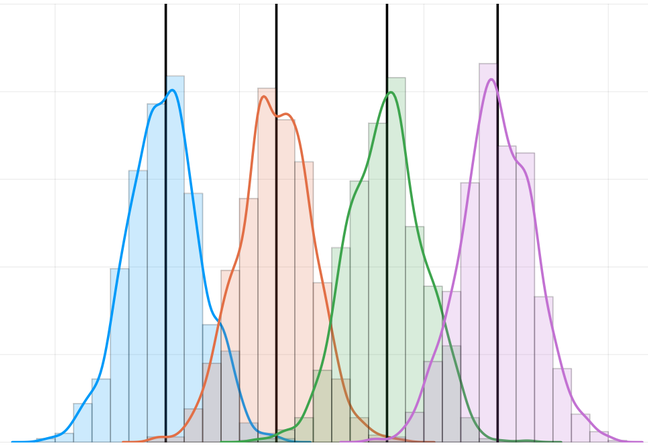
$$Q_i = \frac{1}{R_{i-1}} \sum_{j}^{R_{i-1}} e^{-(\beta_i - \beta_{i-1})E_j}$$



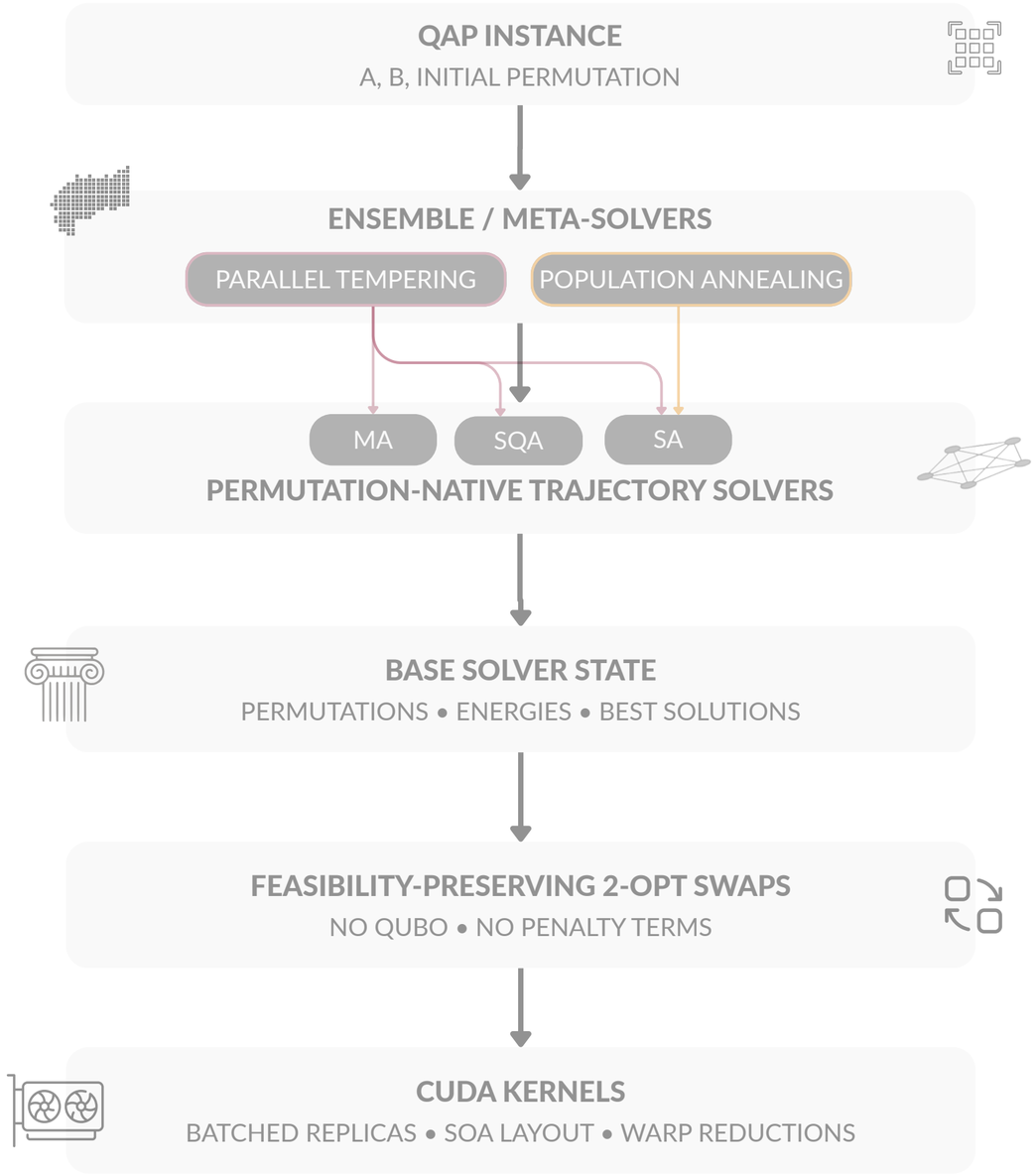
ADAPTIVE TEMPERATURE STEPS

"... uniform effectiveness of resampling is achieved by ensuring a constant overlap of the energy histograms of population members between the neighboring temperatures."

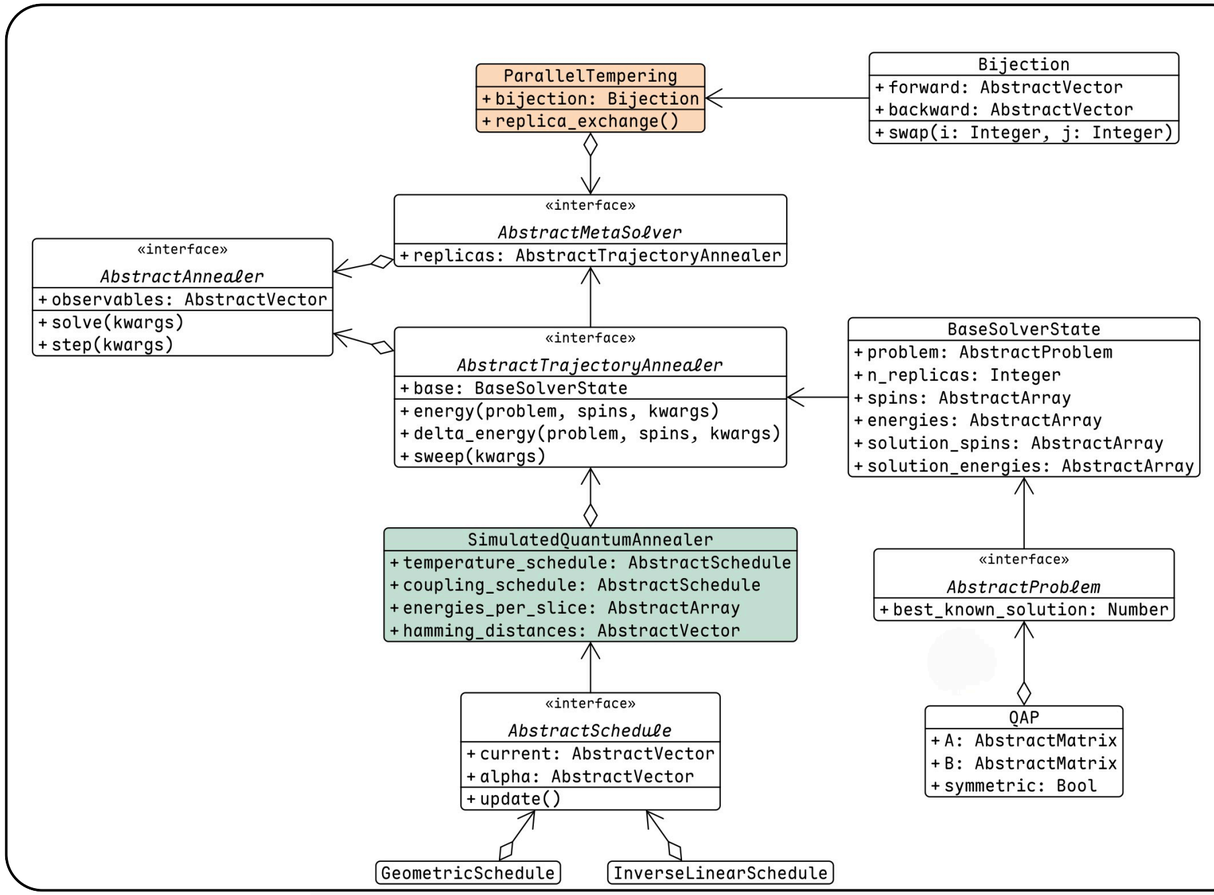
$$\alpha(\beta_{i-1}, \beta') = \frac{1}{R_{i-1}} \sum_{j}^{R_{i-1}} \min\left(1, \frac{Re^{-(\beta\prime - \beta_{i-1})E_j}}{R_{i-1}Q(\beta_{i-1}, \beta')},\right)$$



Composable Framework



ASSA.jl



RESULTS

QAPLIB

Sweep Budgeted

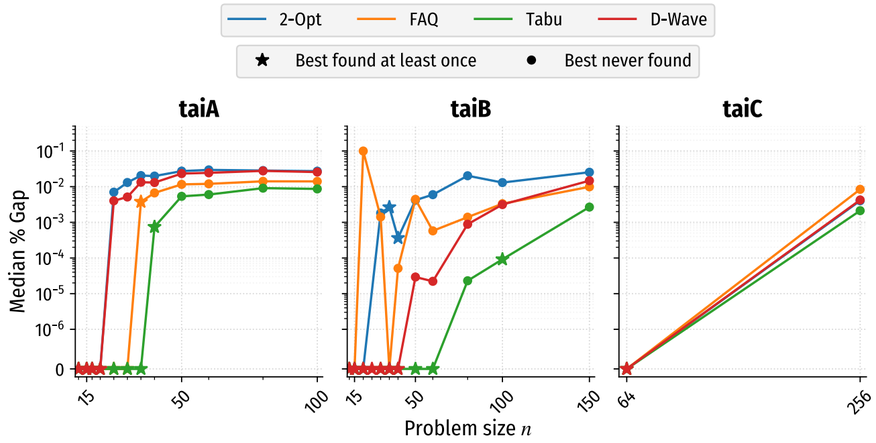
59286 New world record for tai175e found using PA/PTC (previously 59732).

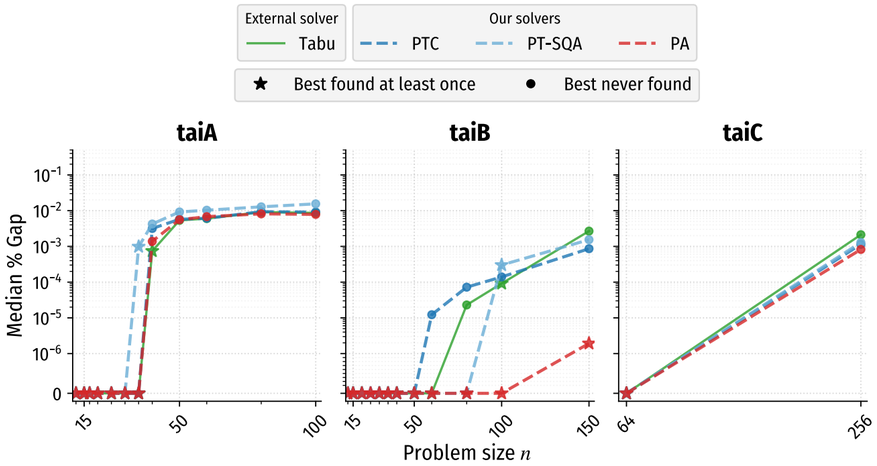
62/150

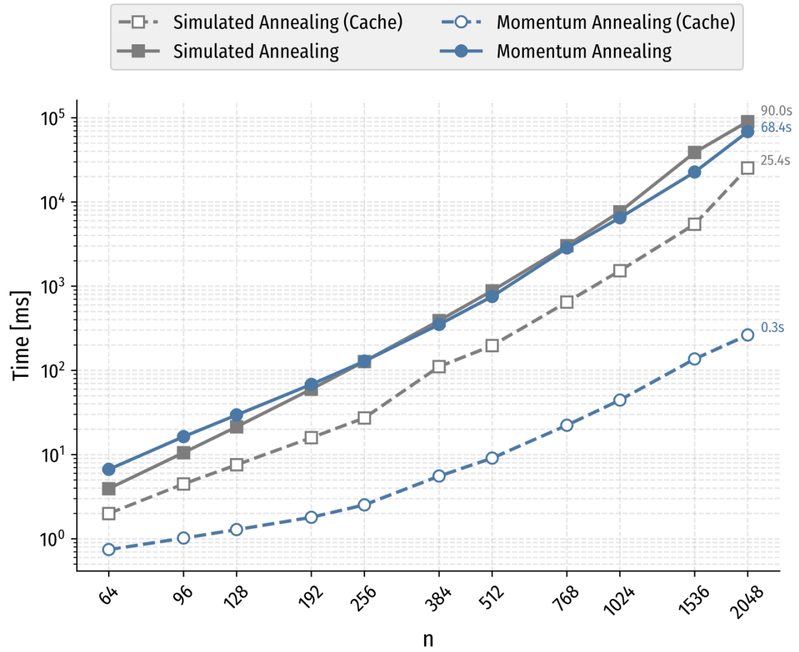
Problem	BKS	SAC			PA			PTC			PT-SQA		
		$\bar{\theta}$	Θ	Time	$\bar{\theta}$	Θ	Time	$\bar{\theta}$	Θ	Time	$\bar{\theta}$	Θ	Time
tai175e01	59732	0.62	0/10	121.0	**−0.20***	**10/10**	38.7	**−0.33***	**10/10**	**21.1**	5.08	0/10	458.3
# of Best		35	21 (302)	8	41	22 (344)	43	40	35 (352)	11	25	17 (295)	0

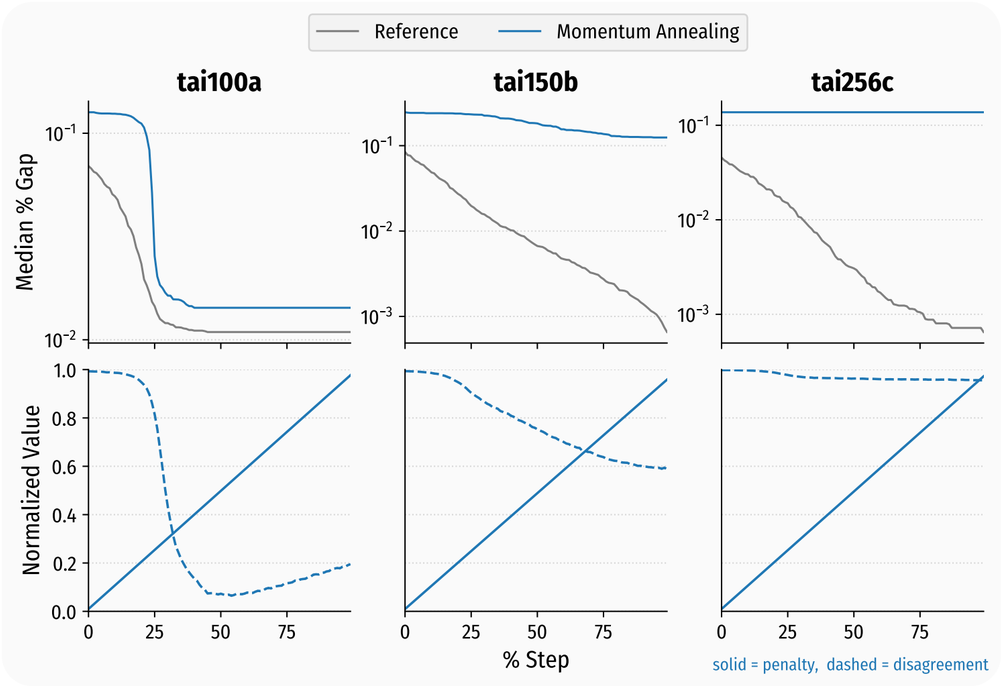
- PA with adaptive temperatures wins
- PT-SQA beats PTC on some instances (taiB)

150s – external







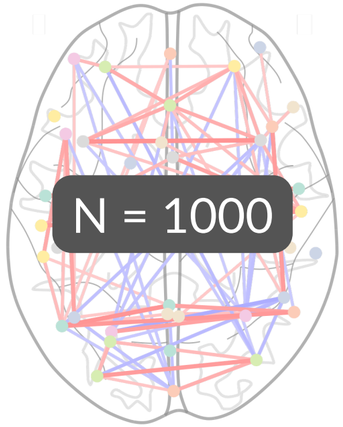


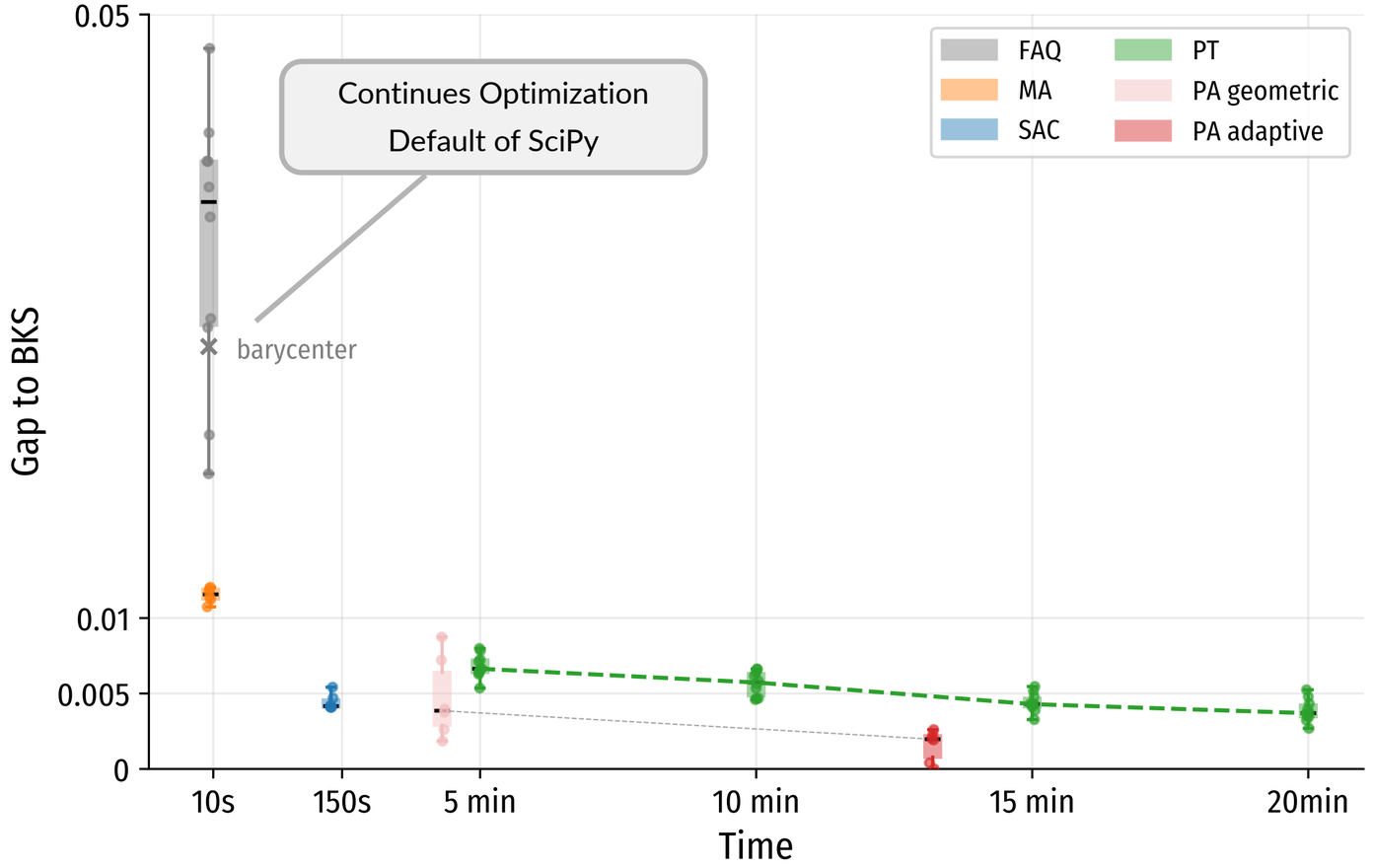


MA, powered by the O(1) cache, sweeps in seconds, which takes SA minutes.

But the solution quality lacks behind in general.







- MA speed-king, PA quality-king

nilearn.datasets.fetch_development_fmri

Examples using nilearn.datasets.fetch_development_fmri: Classification of age groups using functional connectivity Clustering methods to learn a brain parcellation from fMRI Comparing connectomes o...

Nilearn

Schaefer 2018 atlas

Access: See nilearn.datasets.fetch_atlas_schaefer_2018. Notes: This atlas (Schaefer 2018 parcellation 1) provides a labeling of cortical voxels in the MNI152 space, see Schaefer et al. 2. Each ROI...

Nilearn

CONCLUSIONS & FUTURE

01

NATIVE

Spin-glass mechanisms be transferred to integer permutations: Bypass the restrictive QUBO bottleneck entirely by reformulating physics-inspired dynamics to operate natively in integer permutation space.

02

HYBRID

Quantum tunneling and classical thermal fluctuation?
Fused replica exchange with SQA Trotter slicing (PT-SQA).

03

SCALE

High replica parallelism on GPU and designed O(1) cache structures for bipartite QAPs.



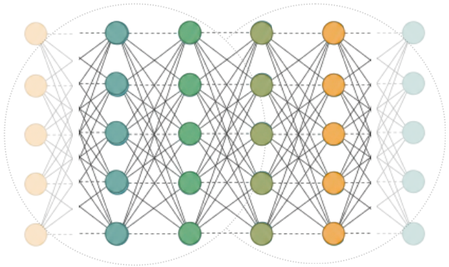
- **new world record**
- scaling to massive **n=1000 human connectomes**



SEEDED GRAPH MATCHING

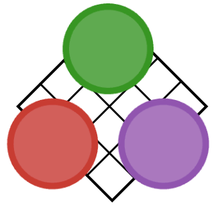
Exploiting known node correspondences by encoding them into **GQAP's** C matrix.

$$\mathrm{tr}(\cdots) + \sum C_{i\pi(i)}$$



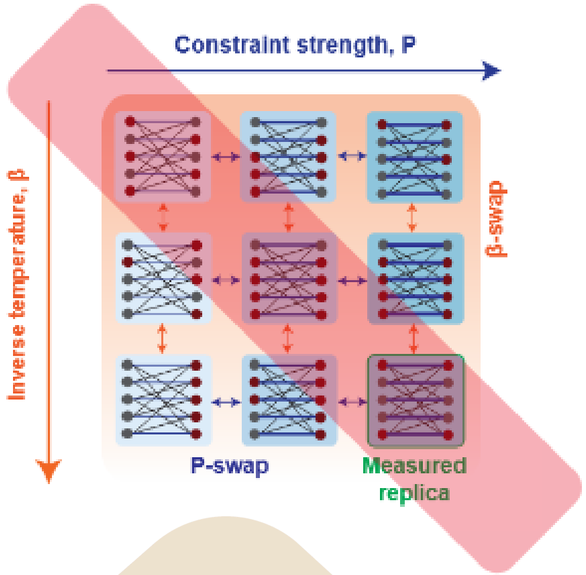
MA-SQA HYBRIDIZATION

Fusing bipartite parallelization architectures with the multi-slice tunneling dynamics of quantum algorithms.

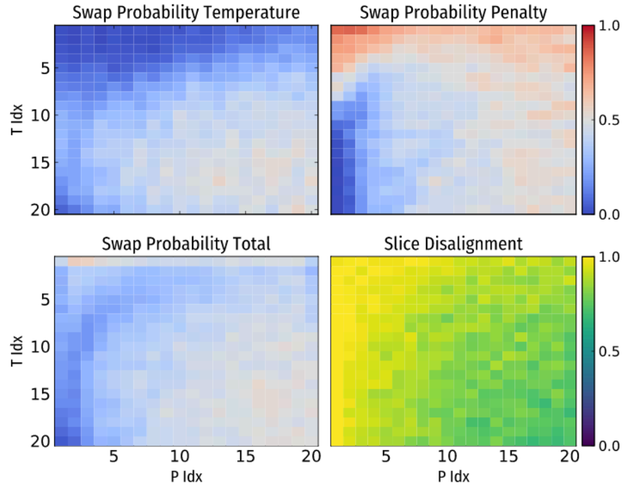


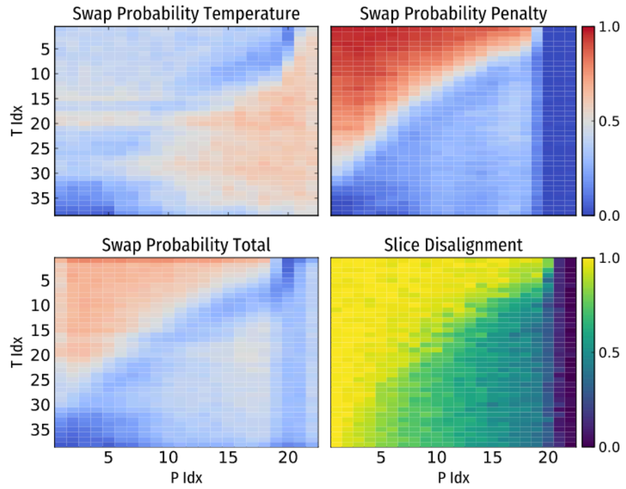
ASSA.jl

Orchestrate solvers; transition to KernelAbstractions.jl; support more (sparse) problem domains; ...

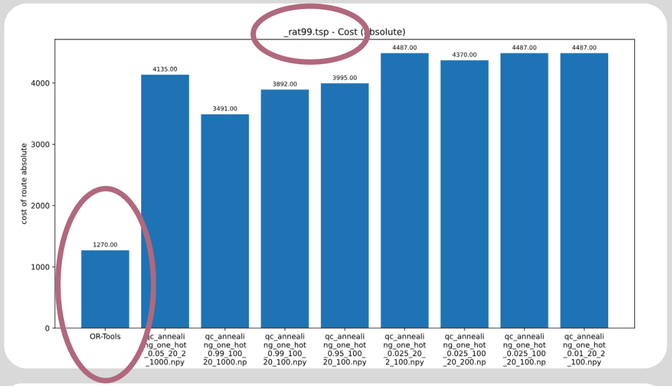

Constraint strength, P
Inverse temperature, β
β-swap
P-swap
Measured replica

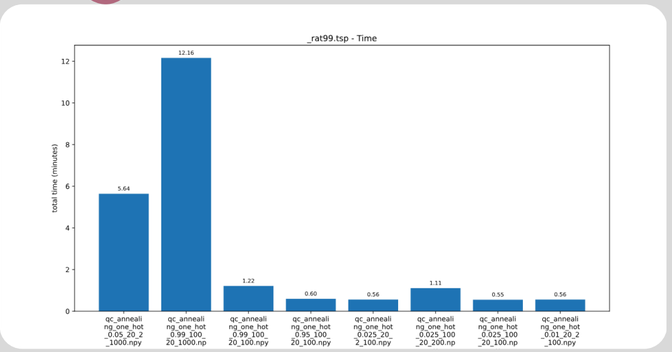

Corentin Delacour · 1:57 AM
Very cool -- thank you for sharing this, you are attacking a very difficult problem! Yes, in simulations we swapped spin states for convenience. Maybe you could try this way to ease the 2D interpretation with static replica indices? Although in hardware, swapping parameters might be more efficient.
About the shuffled penalties: do you mean randomly selecting a penalty parameter?
Cheers


Swap Probability Temperature
Swap Probability Penalty
Swap Probability Total
Slice Disalignment
T Idx
P Idx


Swap Probability Temperature
Swap Probability Penalty
Swap Probability Total
Slice Disalignment
T Idx
P Idx

TSP

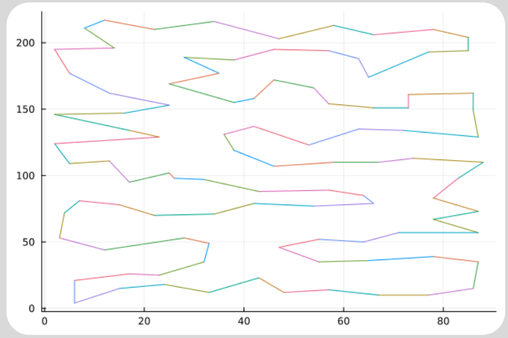




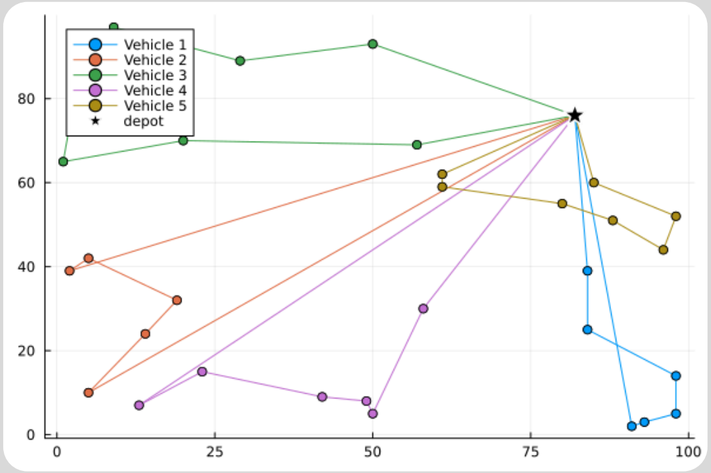
Credits: **@Franz Kasprzyk**

Ours (PA)

```
Progress:  103    Time: 0:00:17
        found: 1215.0    mean: 1219.18
        res: 4.0, gap: 0.003
```



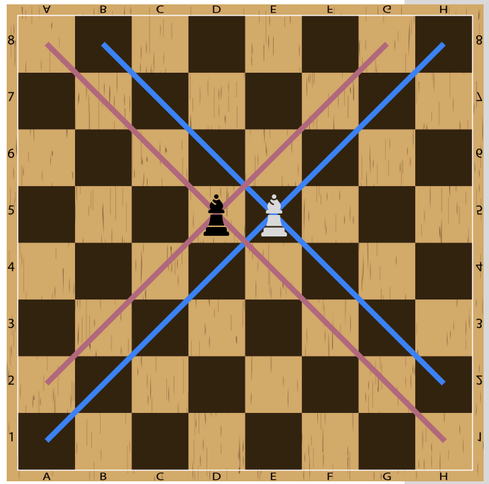
(C)VRP

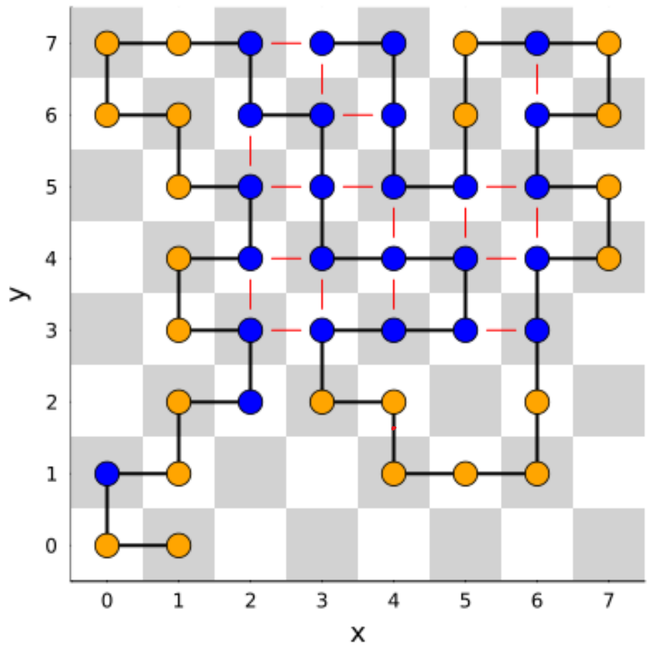


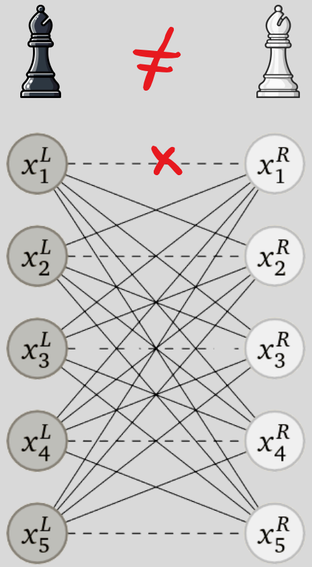

```
slots_per_vehicle
Out[0]: 5×8 Matrix{Int64}:
  1   2   3   4   5   6   7   8
  9  10  11  12  13  14  15  16
 17  18  19  20  21  22  23  24
 25  26  27  28  29  30  31  32
 33  34  35  36  37  38  39  40

sol_perm'
Out[0]: 1×40 adjoint(::Vector{Int64}) with eltype Int64:
 38  31  30  27  23  32  2  12  15  21  11  39  3  35  20  37  7  13  6  24  4  14  29  34  22  36  18  28  19  40  5  1  9  17  25  33  26  10  16  8
```






y
x
0
1
2
3
4
5
6
7


x_1^L x_2^L x_3^L x_4^L x_5^L
x_1^R x_2^R x_3^R x_4^R x_5^R

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